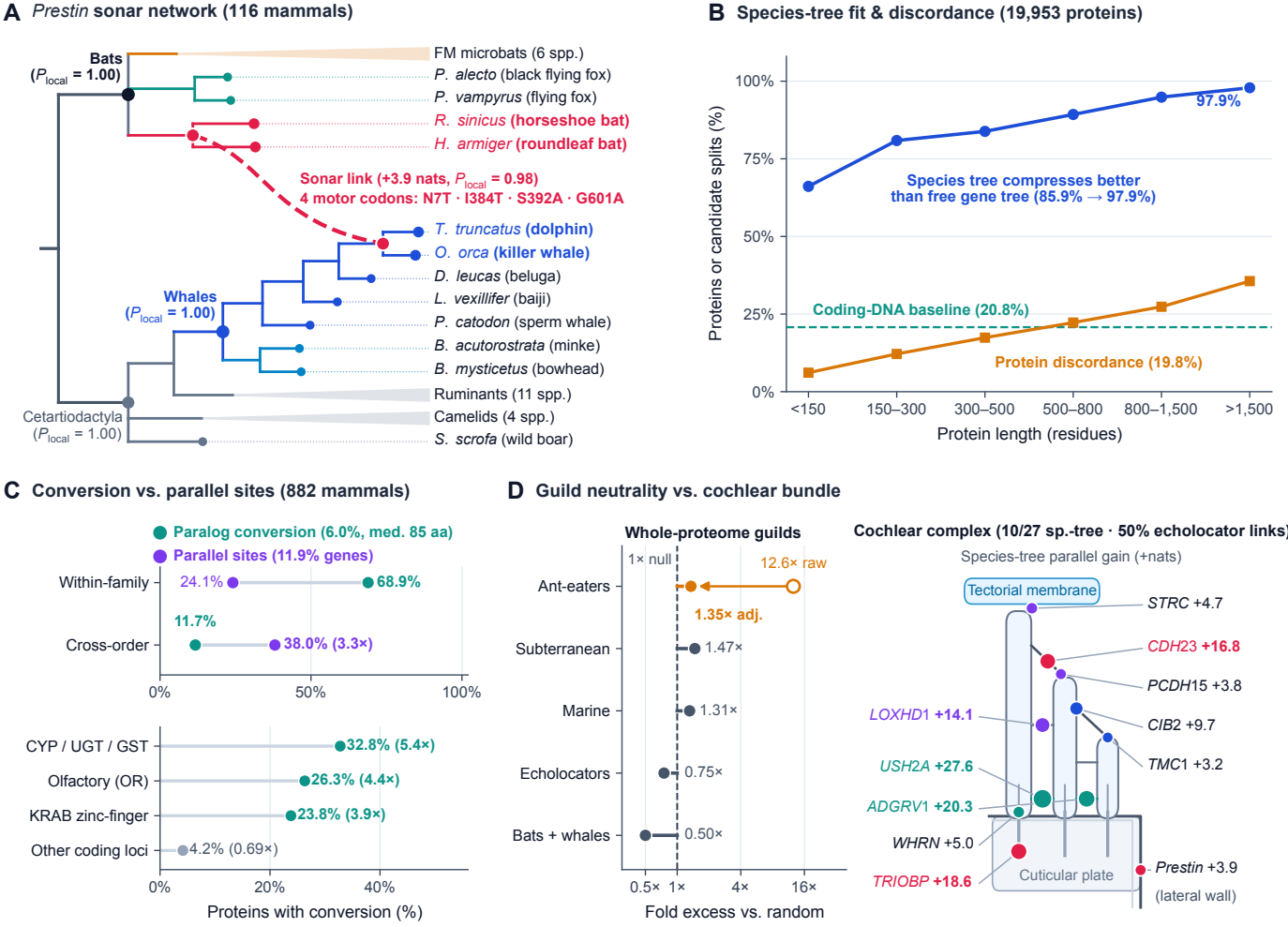

A Prestin sonar network (116 mammals)
Bats
(P local = 1.00)
FM microbats (6 spp.)
P. alecto (black flying fox)
P. vampyrus (flying fox)
R. sinicus (horseshoe bat)
H. armiger (roundleaf bat)
Sonar link (+3.9 nats, P local = 0.98)
4 motor codons: N7T · I384T · S392A · G601A
T. truncatus (dolphin)
O. orca (killer whale)
D. leucas (beluga)
L. vexillifer (baiji)
P. catodon (sperm whale)
B. acutorostrata (minke)
B. mysticetus (bowhead)
Whales
(P local = 1.00)
Ruminants (11 spp.)
Camelids (4 spp.)
S. scrofa (wild boar)
Cetartiodactyla
(P local = 1.00)
B Species-tree fit & discordance (19,953 proteins)
Proteins or candidate splits (%)
100%
75%
50%
25%
0%
97.9%
Species tree compresses better
than free gene tree (85.9% → 97.9%)
Coding-DNA baseline (20.8%)
Protein discordance (19.8%)
<150
150–300
300–500
500–800
800–1,500
>1,500
Protein length (residues)
C Conversion vs. parallel sites (882 mammals)
Paralog conversion (6.0%, med. 85 aa)
Parallel sites (11.9% genes)
Within-family
24.1%
68.9%
Cross-order
11.7%
38.0% (3.3×)
0%
50%
100%
CYP / UGT / GST
32.8% (5.4×)
Olfactory (OR)
26.3% (4.4×)
KRAB zinc-finger
23.8% (3.9×)
Other coding loci
4.2% (0.69×)
0%
20%
40%
Proteins with conversion (%)
D Guild neutrality vs. cochlear bundle
Whole-proteome guilds
1× null
12.6× raw
1.35× adj.
Ant-eaters
Subterranean
1.47×
Marine
1.31×
Echolocators
0.75×
Bats + whales
0.50×
0.5×
1×
4×
16×
Fold excess vs. random
Cochlear complex (10/27 sp.-tree · 50% echolocator links)
Species-tree parallel gain (+nats)
Tectorial membrane
STRC +4.7
CDH23 +16.8
PCDH15 +3.8
LOXHD1 +14.1
CIB2 +9.7
USH2A +27.6
TMC1 +3.2
ADGRV1 +20.3
WHRN +5.0
TRIOBP +18.6
Cuticular plate
Prestin +3.9
(lateral wall)